CS 4650/7650 ECE 4655/7655 Digital Image Processing

Homework 6 – Review of Image Processing Operations [100 pts]

Out: Tu Nov 12 **Fall 2024**
Due: Th Nov 21

Part A Geometric Image Transformations [50 pts]

Geometric image transformations are fundamental in many image processing applications (i.e. image warping, registration, mosaic or panorama generation, dataset augmentation for machine learning, especially deep learning etc.).

Task: Use Python or C/C++ to implement the following image transformations. You can use built-in functions for the transformations (such as cv2.warpAffine, imwarp, cv:: warpAffine).

Write functions to perform the following geometric transformations to the input image. Functions should have the specified input arguments, and should work correctly for any given input (i.e. rotate by any given angle).

1) **Translate(im,tx,ty):** Translate input image Im by (tx, ty) pixels **(5 points)**
2) **CropScale(im, x1,y1,x2,y2,s):** Crop a rectangular region from the input image and scale (resize) the cropped region by scale s. (x1,y1) and (x2,y2) define the upper left and lower right coordinates of the rectangle. **(15 points)**
3) **Vertical_Flip(im):** Flip image about y-axis **(5 points)**
4) **Horizontal_Flip(im):** Flip about x-axis **(5 points)**
5) **Rotate(im, angle):** Rotate by a given angles between -180 deg to +180 deg **(15 points)**
6) **Fill(im, x1,y1,x2,y2,val):** Fill rectangular regions defined by the upper left and lower right coordinates (x1,y1) and (x2,y2) with intensity value val **(5 points)**

Report: Apply the above transformations to the given test image. Include output images to the report. Include output image files to submission file.
Test parameters:

(1) tx=300, ty=200;
(2) x1=500, y1=1,x2=1000, y2=800, s=0.5 (half size in x and y);
(5) angle=60;
(6) x1=500, y1=1, x2=1000, y2=800, val=150.

Part B Edge Detection and Histogram of Gradient Orientations [50 pts]

Edges and edge orientations are fundamental for many image processing applications including image segmentation and object detection or recognition. As part of this assignment, you will analyze the image gradients at two scales:

(1) a small scale to detect fine details in the input image;
(2) a larger scale to detect prominent structures in the image.

Use Python or C/C++ to perform the following tasks. You can use built-in functions in Python for image filtering, convolution, and histogram computation.

Tasks:

1) Compute and display Ix, image gradient in x
2) Compute and display Iy, image gradient in y
3) Compute gradient magnitude M
4) Compute gradient orientation theta
5) Compute histogram of gradient orientations H, for bins 0 degree to 359 degree with step size 1 degree.

Report: for two scales, display in the report Ix, Iy, M images and plot histogram H.

Note: Compute the orientation of the edge response by computing the angle theta of the edge gradient where **theta=** $\mathbf{tan^{-1}(Iy/Ix)}$. Note that to get the correct angle between 0 and 360 degrees, where zero is the x-axis, you should use the **atan2(Ix,Iy)** function. Note that you will need to convert from radians to degrees and add 360 if the angle is negative, to change the range of the atan2 function from (-Pi, +Pi] to (0,360] degree in the counter clockwise (CCW) direction.

Submission instructions: Your submission should include a report (including output images & your interpretation of the outputs) and associated programs (as separate files).

References:

[1] Slides for Lecture 12 on Canvas: Lec12_GeometricTransforms.pdf
[2] Szeliski, Computer Vision Book, Chapter 3.6. (https://szeliski.org/Book/)
[3] Gonzalez & Woods, Digital Image Processing, Chapter 2
[4] Slides for Lecture 8 on Canvas: Lec8_EdgeDetection.pdf